

Term 3 Practice Activity: C7 Inverse Normal Distribution

Contents

- 1 Top 5% Exam Scores
- 2 Bottom 10% Commute Times
- 3 Central 95% Device Battery Life
- 4 Bottom 20% Streaming Speeds
- 5 Top 2% Plant Heights
- 6 Bottom 1% Package Delivery Times
- 7 Middle 80% Fitness Run Times
- 8 Top 10% Product Lifetime
- 9 Middle 90% Manufacturing Tolerance Width
- 10 Middle 96% Chemistry Scores

Evaluated criteria

- C7. Uses inverse normal reasoning to find values and intervals.

Standard normal table (values of $\Phi(z)$). Use this same table for all problems in this section.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

1 Top 5% Exam Scores

Problem. A large exam has scores modeled by

$$X \sim N(72, 9^2).$$

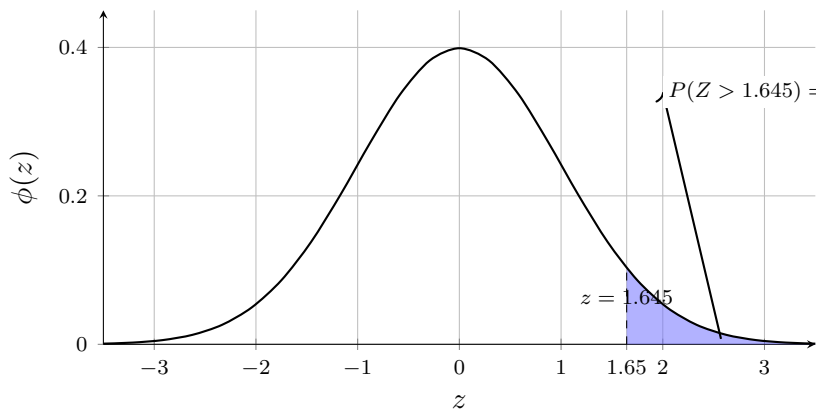
Questions/Tasks. Identify the target probability before using inverse normal, then find the score x that marks the top 5% of students.

C7

Top 5% means

$$P(X > x) = 0.05 \quad \longrightarrow \quad P(X \leq x) = 0.95.$$

$$P(Z > 1.645) = 0.05.$$



Standardize:

$$P\left(Z \leq \frac{x - 72}{9}\right) = 0.95.$$

$$P\left(Z \leq \frac{x - 72}{9}\right) = 0.95 \quad \longrightarrow \quad z_{0.95} \approx 1.645 \quad \longrightarrow \quad \frac{x - 72}{9} = 1.645$$

$$\frac{x - 72}{9} = 1.645 \quad \longrightarrow \quad x - 72 = 1.645(9) \quad \longrightarrow \quad x = 72 + 1.645(9) \quad \longrightarrow \quad x = 86.805.$$

$$x \approx 86.8$$

Interpretation. A score of about 86.8 is the cutoff so only about 5% of students score higher.

2 Bottom 10% Commute Times

Problem. Morning commute times (minutes) are modeled by

$$X \sim N(38, 7^2).$$

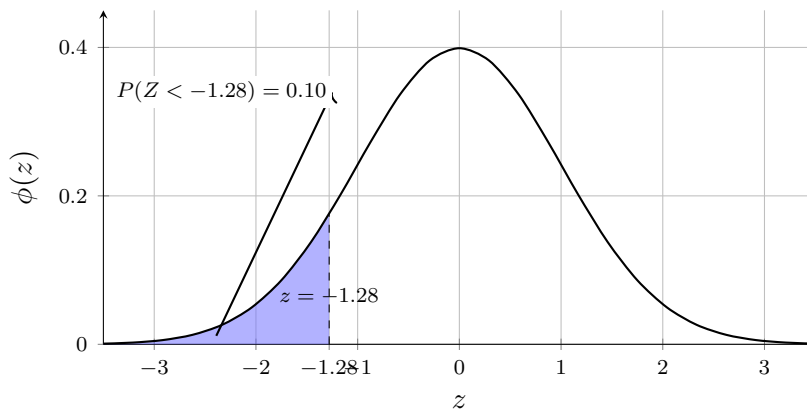
Questions/Tasks. Identify the target probability before using inverse normal, then find the time x below which the bottom 10% of commutes fall.

C7

Bottom 10% means

$$P(X \leq x) = 0.10.$$

$$P(Z < -1.28) = 0.10.$$



Convert to standard normal:

$$P\left(Z \leq \frac{x - 38}{7}\right) = 0.10.$$

$$P\left(Z \leq \frac{x - 38}{7}\right) = 0.10 \quad \longrightarrow \quad z_{0.10} \approx -1.28 \quad \longrightarrow \quad \frac{x - 38}{7} = -1.28$$

$$\frac{x - 38}{7} = -1.28 \quad \longrightarrow \quad x - 38 = -1.28(7) \quad \longrightarrow \quad x = 38 - 1.28(7) \quad \longrightarrow \quad x = 29.04.$$

$$x \approx 29.0 \text{ min}$$

Interpretation. About 10% of commutes are 29.0 minutes or shorter.

3 Central 95% Device Battery Life

Problem. Battery life (hours) for a device model follows

$$X \sim N(20, 2.5^2).$$

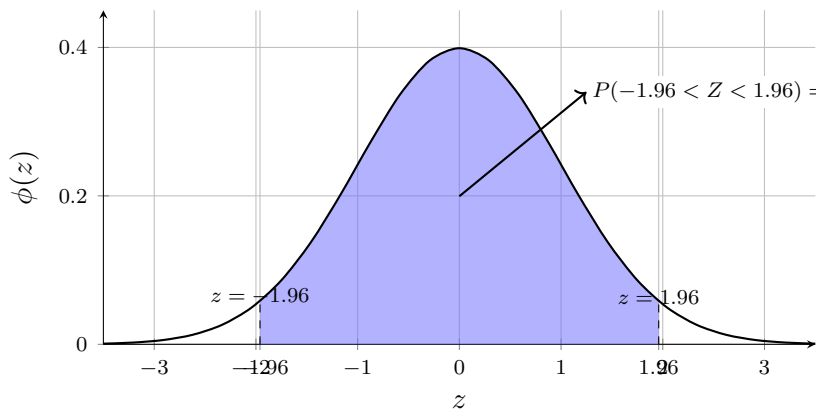
Questions/Tasks. Identify the target probabilities before using inverse normal, then find the interval $[a, b]$ containing the middle 95% of battery lives.

C7

Middle 95% leaves 2.5% in each tail:

$$P(X \leq a) = 0.025, \quad P(X \leq b) = 0.975.$$

$$P(-1.96 < Z < 1.96) = 0.95.$$



$$P(X \leq a) = 0.025 \quad \longrightarrow \quad z_{0.025} \approx -1.96 \quad \longrightarrow \quad \frac{a - 20}{2.5} = -1.96$$

$$\longrightarrow \quad a = 20 - 1.96(2.5) \quad \longrightarrow \quad a = 15.10.$$

$$P(X \leq b) = 0.975 \quad \longrightarrow \quad z_{0.975} \approx 1.96 \quad \longrightarrow \quad \frac{b - 20}{2.5} = 1.96$$

$$\longrightarrow \quad b = 20 + 1.96(2.5) \quad \longrightarrow \quad b = 24.90.$$

$$[a, b] \approx [15.1, 24.9] \text{ h}$$

Interpretation. About 95% of batteries last between 15.1 h and 24.9 h.

4 Bottom 20% Streaming Speeds

Problem. Home streaming speed (Mbps) is modeled by

$$X \sim N(85, 12^2).$$

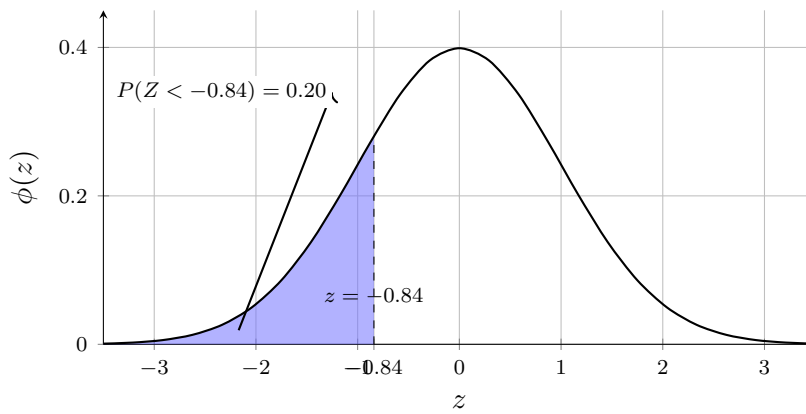
Questions/Tasks. Identify the target probability before using inverse normal, then find the speed x that separates the bottom 20% of connections.

C7

Bottom 20% means

$$P(X \leq x) = 0.20.$$

$$P(Z < -0.84) = 0.20.$$



Convert to Z :

$$P\left(Z \leq \frac{x - 85}{12}\right) = 0.20.$$

$$P\left(Z \leq \frac{x - 85}{12}\right) = 0.20 \quad \longrightarrow \quad z_{0.20} \approx -0.84 \quad \longrightarrow \quad \frac{x - 85}{12} = -0.84$$

$$\frac{x - 85}{12} = -0.84 \quad \longrightarrow \quad x - 85 = -0.84(12) \quad \longrightarrow \quad x = 85 - 0.84(12) \quad \longrightarrow \quad x = 74.92.$$

$$x \approx 74.9 \text{ Mbps}$$

Interpretation. Roughly 20% of homes have streaming speed below about 74.9 Mbps.

5 Top 2% Plant Heights

Problem. The height of plants in a greenhouse (cm) follows

$$X \sim N(48, 6^2).$$

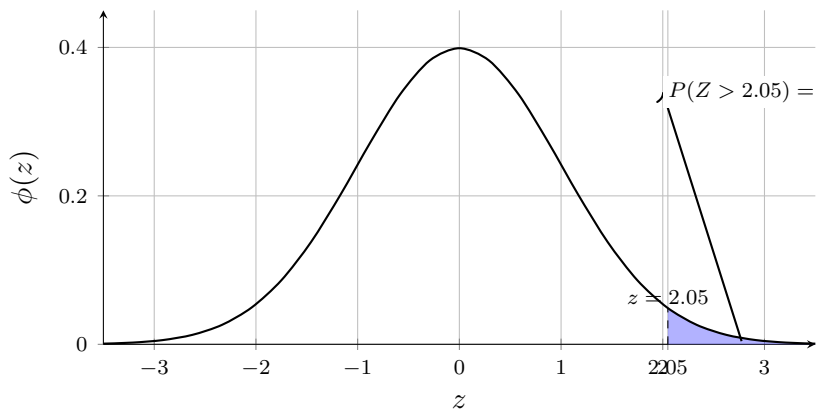
Questions/Tasks. Identify the target probability before using inverse normal, then find the height x so that only the tallest 2% are above it.

C7

Top 2% means

$$P(X > x) = 0.02 \quad \longrightarrow \quad P(X \leq x) = 0.98.$$

$$P(Z > 2.05) = 0.02.$$



Standardize:

$$P\left(Z \leq \frac{x - 48}{6}\right) = 0.98.$$

$$P\left(Z \leq \frac{x - 48}{6}\right) = 0.98 \quad \longrightarrow \quad z_{0.98} \approx 2.05 \quad \longrightarrow \quad \frac{x - 48}{6} = 2.05$$

$$\frac{x - 48}{6} = 2.05 \quad \longrightarrow \quad x - 48 = 2.05(6) \quad \longrightarrow \quad x = 48 + 2.05(6) \quad \longrightarrow \quad x = 60.30.$$

$$x \approx 60.3 \text{ cm}$$

Interpretation. Plants taller than about 60.3 cm are in the tallest 2%.

6 Bottom 1% Package Delivery Times

Problem. Delivery times for a service (hours) are modeled by

$$X \sim N(36, 5^2).$$

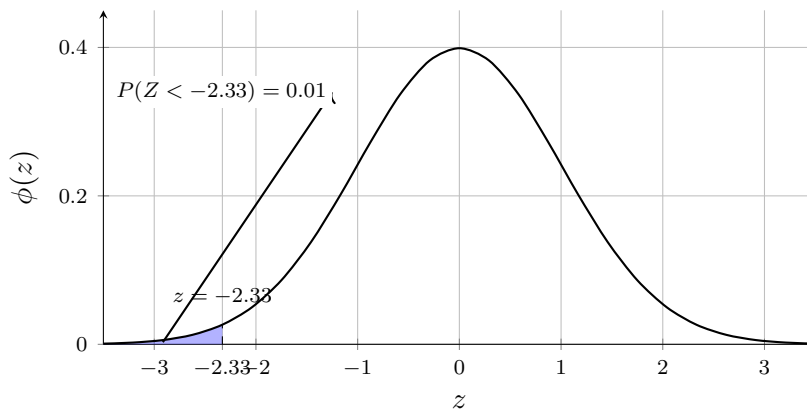
Questions/Tasks. Identify the target probability before using inverse normal, then find the delivery time x below which the fastest 1% of deliveries occur.

C7

Bottom 1% means

$$P(X \leq x) = 0.01.$$

$$P(Z < -2.33) = 0.01.$$



Convert to a standard normal statement:

$$P\left(Z \leq \frac{x - 36}{5}\right) = 0.01.$$

$$P\left(Z \leq \frac{x - 36}{5}\right) = 0.01 \quad \longrightarrow \quad z_{0.01} \approx -2.33 \quad \longrightarrow \quad \frac{x - 36}{5} = -2.33$$

$$\frac{x - 36}{5} = -2.33 \quad \longrightarrow \quad x - 36 = -2.33(5) \quad \longrightarrow \quad x = 36 - 2.33(5) \quad \longrightarrow \quad x = 24.35.$$

$$x \approx 24.4 \text{ h}$$

Interpretation. About 1% of packages are delivered in 24.4 hours or less.

7 Middle 80% Fitness Run Times

Problem. A 2 km run time (minutes) for students is modeled by

$$X \sim N(11.8, 1.4^2).$$

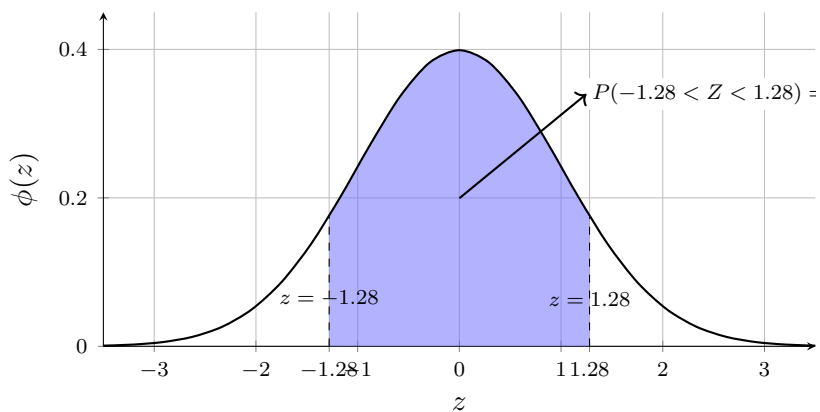
Questions/Tasks. Identify the target probabilities before using inverse normal, then find the interval $[a, b]$ containing the middle 80% of run times.

C7

Middle 80% leaves 10% in each tail:

$$P(X \leq a) = 0.10, \quad P(X \leq b) = 0.90.$$

$$P(-1.28 < Z < 1.28) = 0.80.$$



$$\begin{aligned}
 P(X \leq a) = 0.10 & \longrightarrow z_{0.10} \approx -1.28 & \longrightarrow \frac{a - 11.8}{1.4} = -1.28 \\
 & \longrightarrow a = 11.8 - 1.28(1.4) & \longrightarrow a = 10.008. \\
 P(X \leq b) = 0.90 & \longrightarrow z_{0.90} \approx 1.28 & \longrightarrow \frac{b - 11.8}{1.4} = 1.28 \\
 & \longrightarrow b = 11.8 + 1.28(1.4) & \longrightarrow b = 13.592.
 \end{aligned}$$

$$[a, b] \approx [10.0, 13.6] \text{ min}$$

Interpretation. About 80% of students finish the 2 km run between 10.0 and 13.6 minutes.

8 Top 10% Product Lifetime

Problem. A light bulb lifetime (hours) is modeled by

$$X \sim N(1200, 150^2).$$

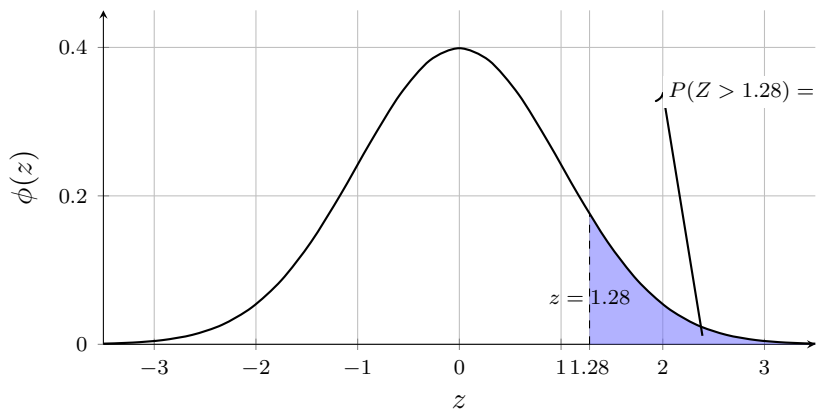
Questions/Tasks. Identify the target probability before using inverse normal, then find the lifetime x that marks the top 10% longest-lasting bulbs.

C7

Top 10% means

$$P(X > x) = 0.10 \longrightarrow P(X \leq x) = 0.90.$$

$$P(Z > 1.28) = 0.10.$$



Convert to standard normal:

$$P\left(Z \leq \frac{x - 1200}{150}\right) = 0.90.$$

$$P\left(Z \leq \frac{x - 1200}{150}\right) = 0.90 \quad \longrightarrow \quad z_{0.90} \approx 1.28 \quad \longrightarrow \quad \frac{x - 1200}{150} = 1.28$$

$$\frac{x - 1200}{150} = 1.28 \quad \longrightarrow \quad x - 1200 = 1.28(150) \quad \longrightarrow \quad x = 1200 + 1.28(150) \quad \longrightarrow \quad x = 1392.$$

$$x \approx 1392 \text{ h}$$

Interpretation. Only about 10% of bulbs last longer than roughly 1392 hours.

9 Middle 90% Manufacturing Tolerance Width

Problem. A metal rod diameter (mm) is modeled by

$$X \sim N(10.00, 0.08^2).$$

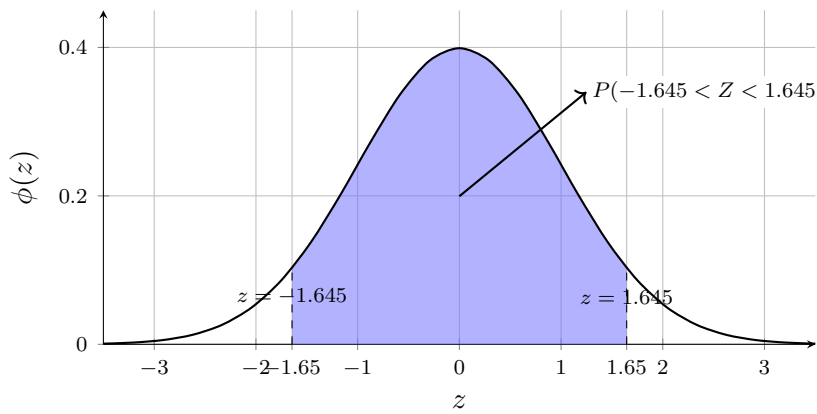
Questions/Tasks. Identify the target probabilities before using inverse normal, then find the central 90% interval $[a, b]$ for acceptable diameters.

C7

Middle 90% leaves 5% in each tail:

$$P(X \leq a) = 0.05, \quad P(X \leq b) = 0.95.$$

$$P(-1.645 < Z < 1.645) = 0.90.$$



$$\begin{aligned}
 P(X \leq a) = 0.05 &\longrightarrow z_{0.05} \approx -1.645 \longrightarrow \frac{a - 10.00}{0.08} = -1.645 \\
 &\longrightarrow a = 10.00 - 1.645(0.08) \longrightarrow a = 9.8684. \\
 P(X \leq b) = 0.95 &\longrightarrow z_{0.95} \approx 1.645 \longrightarrow \frac{b - 10.00}{0.08} = 1.645 \\
 &\longrightarrow b = 10.00 + 1.645(0.08) \longrightarrow b = 10.1316.
 \end{aligned}$$

$$[a, b] \approx [9.868, 10.132] \text{ mm}$$

Interpretation. About 90% of rod diameters are expected between 9.868 mm and 10.132 mm.

10 Middle 96% Chemistry Scores

Problem. Chemistry quiz scores are modeled by

$$X \sim N(64, 10^2).$$

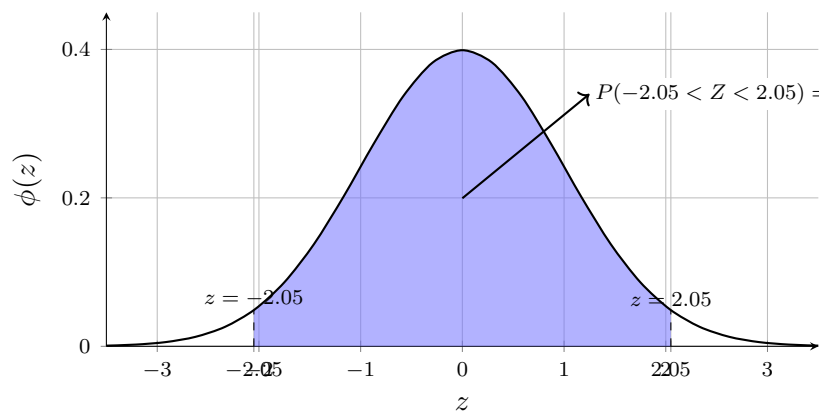
Questions/Tasks. Identify the target probabilities before using inverse normal, then find the score interval $[a, b]$ containing the middle 96% of students.

C7

Middle 96% leaves 2% in each tail:

$$P(X \leq a) = 0.02, \quad P(X \leq b) = 0.98.$$

$$P(-2.05 < Z < 2.05) = 0.96.$$



$$\begin{aligned}
 P(X \leq a) = 0.02 &\longrightarrow z_{0.02} \approx -2.05 &\longrightarrow \frac{a - 64}{10} = -2.05 \\
 &\longrightarrow a = 64 - 2.05(10) &\longrightarrow a = 43.5. \\
 P(X \leq b) = 0.98 &\longrightarrow z_{0.98} \approx 2.05 &\longrightarrow \frac{b - 64}{10} = 2.05 \\
 &\longrightarrow b = 64 + 2.05(10) &\longrightarrow b = 84.5.
 \end{aligned}$$

$$[a, b] \approx [43.5, 84.5]$$

Interpretation. About 96% of students score between roughly 43.5 and 84.5 points.